

Title of EGH445 Assessment/Report

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Abstract—This report outlines the comprehensive modelling, discretisation, and control system design for an underactuated Remotely Operated Vehicle (ROV) operating in a complex hydrodynamic environment. A 3-DOF planar model is derived from the full 6-DOF rigid-body dynamics and linearised around a stable equilibrium point. A discrete-time Linear Quadratic Integral (LQI) controller featuring an advanced anti-windup scheme is implemented. The LQI design process is detailed, demonstrating robust tracking on linear, non-linear, and highly noisy blackbox plants despite severe actuator saturation constraints. Furthermore, a discrete-time Luenberger observer is designed using optimal pole placement techniques for the dead-reckoning of unmeasured states (x, z). Extensive simulation analysis demonstrates that while the LQI controller is highly effective at disturbance rejection, the observer inevitably drifts over time due to noise integration. The report concludes that dead reckoning is highly discouraged for this type of system, emphasizing the necessity of absolute positioning sensors for autonomous subsea operations.

I. INTRODUCTION

This report analyzes the modeling and control strategies of an underactuated research ROV to achieve optimal performance and autonomous waypoint tracking. Operating in complex hydrodynamic environments, this study leverages Fossen’s framework [1] and SNAME standards [2] for state-space representation, alongside Scilletta’s work [3] for parameter identification. The focus is on a specific variant (similar to Fig. 1) featuring translated aft horizontal thrusters and 90-degree rotated vertical thrusters. The objective is to design a robust output-feedback controller capable of navigating this vessel accurately, while critically evaluating the performance limitations and engineering trade-offs inherent in the subsea domain.

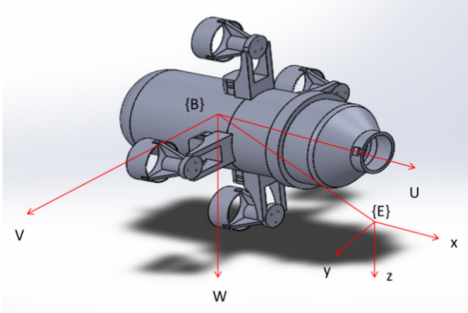


Fig. 1. Example of an underactuated research ROV. Image from [4].

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II. SYSTEM MODELING

Applying an Earth-fixed inertial frame $\{E\}$ and a Body-fixed frame $\{B\}$, system states are defined as position/orientation $\eta = [x, y, z, \phi, \theta, \psi]^T$ and body velocities $\nu = [u, v, w, p, q, r]^T$. The 6-DOF rigid-body equations of motion are given by standard marine hydrodynamic theory [1]:

$$\dot{\eta} = J(\eta)\nu, \quad M\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau \quad (1)$$

A. Decoupling and Design Parameters

Assuming low-speed operation in calm water with negligible roll, sway, and yaw, the dynamics safely decouple into a 3-DOF longitudinal planar model (surge x, u , heave z, w , pitch θ, q). This targeted planar representation is vital as it heavily streamlines control derivation while preserving the primary operational workspace of the ROV. Relevant mass, inertia, and hydrodynamic parameters are updated for the specific dual-horizontal/vertical thruster configuration (Table I).

TABLE I
ROV PLANAR MODEL PARAMETERS

Sym.	Value	Unit	Sym.	Value	Unit
m	16.6	kg	X_u	9.716	N s/m
W	162.846	N	Z_w	68.855	N s/m
B	168.928	N	M_q	0.07	N s/rad
I_y	0.66054	kg m ²	$X_{u u }$	18.18	N s ² /m ²
$X_{\dot{u}}$	1.3169	kg	$Z_{w w }$	36.99	N s ² /m ²
$Z_{\dot{w}}$	14.4260	kg	$M_{q q }$	1.55	N s ² /rad ²
$M_{\dot{q}}$	0.3637	kg m ² /rad	z_g	0.03	m
$l_{x,v}$	0.0	m	$l_{z,h,top}$	-0.1	m
$l_{z,h,bot}$	0.1	m			

The resulting 3-DOF conceptually derived matrices isolate the coupled dynamics into distinct translation and rotational components:

$$M = \text{diag}(m - X_{\dot{u}}, m - Z_{\dot{w}}, I_y - M_{\dot{q}}) \quad (2)$$

$$D(\nu) = -\text{diag}(X_u + X_{u|u|}|u|, Z_w + Z_{w|w|}|w|, M_q + M_{q|q|}|q|) \quad (3)$$

$$g(\eta) = [0, 0, z_g W \sin(\theta)]^T \quad (4)$$

Notably, the hydrostatic restoring forces $g(\eta)$ provide a natural, passive pendulum effect that fights pitch disturbances, a crucial element for intrinsic stability.

B. State Space Representation

This decoupling reduces the state space to a 3 DOF system with states $\mathbf{x} = [x, z, \theta, u, w, q]^T$ and control inputs $\mathbf{u}_c = [u_1, u_2, u_3, u_4]^T$. The continuous non-linear state space model takes the State-Dependent Coefficient (SDC) form:

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x})\mathbf{x} + \mathbf{B}\mathbf{u}_c \quad (5)$$

The non-linear state-dependent system matrix $\mathbf{A}(\mathbf{x})$ captures the full complexity of the planar kinematics, quadratic drag, and coupled Coriolis forces:

$$\mathbf{A}(\mathbf{x}) = \begin{bmatrix} 0 & 0 & 0 & \cos \theta & \sin \theta & 0 \\ 0 & 0 & 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & a_{44}(u) & a_{45}(q) & 0 \\ 0 & 0 & 0 & a_{54}(q) & a_{55}(w) & 0 \\ 0 & 0 & a_{63}(\theta) & a_{64}(w) & 0 & a_{66}(q) \end{bmatrix} \quad (6)$$

The embedded non-linear coefficients are defined as:

- Surge Damping: $a_{44}(u) = -(X_u + X_{u|u}|u|)/(m - X_{\dot{u}})$
- Surge Coriolis: $a_{45}(q) = -(m - Z_{\dot{w}})q/(m - X_{\dot{u}})$
- Heave Coriolis: $a_{54}(q) = (m - X_{\dot{u}})q/(m - Z_{\dot{w}})$
- Heave Damping: $a_{55}(w) = -(Z_w + Z_{w|w}|w|)/(m - Z_{\dot{w}})$
- Pitch Restoring: $a_{63}(\theta) = \frac{-z_g W}{I_y - M_{\dot{q}}} \left(\frac{\sin \theta}{\theta} \right)$
- Pitch Munk Moment: $a_{64}(w) = (Z_{\dot{w}} - X_{\dot{u}})w/(I_y - M_{\dot{q}})$
- Pitch Damping: $a_{66}(q) = -(M_q + M_{q|q}|q|)/(I_y - M_{\dot{q}})$

The input matrix $\mathbf{B}$ is modified to reflect the physical geometry of the split thrusters. Because the vertical thrusters (u_3, u_4) are located dead-center longitudinally ($l_{x,v} = 0$), they exert strictly zero pitch moment, placing the entire burden of active pitch authority on the stacked horizontal thrusters (u_1, u_2):

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ b_{41} & b_{41} & 0 & 0 \\ 0 & 0 & b_{53} & b_{53} \\ b_{61,t} & b_{61,b} & 0 & 0 \end{bmatrix} \quad (7)$$

where $b_{41} = \frac{1}{m - X_{\dot{u}}}$, $b_{53} = \frac{1}{m - Z_{\dot{w}}}$, $b_{61,t} = \frac{l_{z,h,top}}{I_y - M_{\dot{q}}}$, and $b_{61,b} = \frac{l_{z,h,bot}}{I_y - M_{\dot{q}}}$.

C. Linearisation and Equilibrium Justification

To apply modern linear control theory, the system must be linearised. An equilibrium point representing a stable hover was selected, where all velocities and orientation angles are zero ($\theta = 0, u = 0, w = 0, q = 0$).

A critical engineering insight is that the spatial coordinates x and z are translationally invariant in calm water. The gravitational and buoyant forces acting on the vehicle do not change regardless of its absolute position in the fluid domain. Therefore, x and z can take arbitrary values at equilibrium. By selecting the origin $\mathbf{x}_e = \mathbf{0}, \mathbf{u}_{c,e} = \mathbf{0}$ as the linearisation point, the resulting Jacobian matrices apply universally across the entire spatial workspace.

The linearised $\mathbf{A}$ matrix simplifies drastically. Because linearisation is evaluated at zero velocity, all non-linear quadratic hydrodynamic damping and Coriolis terms mathematically collapse. Furthermore, small-angle approximations are applied to the gravity restoring moments ($\sin(\theta) \approx \theta$):

$$\mathbf{A}_{lin} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \frac{-X_u}{m - X_{\dot{u}}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{-Z_w}{m - Z_{\dot{w}}} & 0 \\ 0 & 0 & \frac{-z_g W}{I_y - M_{\dot{q}}} & 0 & 0 & \frac{-M_q}{I_y - M_{\dot{q}}} \end{bmatrix} \quad (8)$$

Because the thruster configuration and rigid-body geometry are entirely independent of the spatial orientation near hover, the linearised input matrix $\mathbf{B}_{lin}$ maps identically to the non-linear system:

$$\mathbf{B}_{lin} = \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ b_{41} & b_{41} & 0 & 0 \\ 0 & 0 & b_{53} & b_{53} \\ b_{61,t} & b_{61,b} & 0 & 0 \end{bmatrix} \quad (9)$$

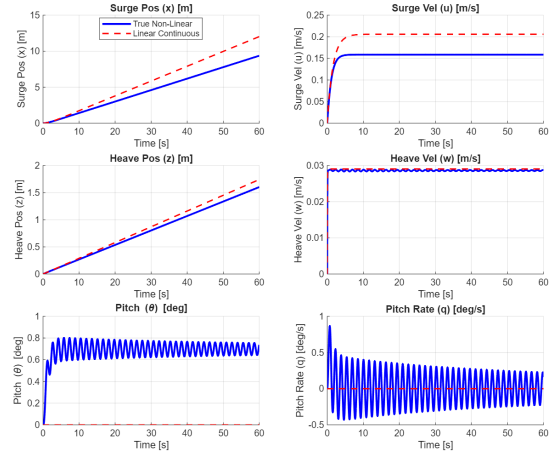


Fig. 2. Comparison of Linear and Non-Linear model depth and pitch responses to an amplified step input.

The profound effect of linearisation is evident in the open-loop response (Fig. 2). The absence of quadratic damping in the linear model causes it to predict a slightly faster, less damped response. Furthermore, the linear model completely fails to capture the sustained oscillatory pitch motion caused by the non-linear Munk moment, which is highly visible in the true non-linear response. This glaring unmodelled pitch variation directly drives the engineering selection of an LQI controller, as standard proportional feedback would fail to reject these persistent non-linear oscillatory disturbances without steady-state error. Despite this discrepancy, the linear model accurately captures the fundamental coupled dynamics and dominant time constants, confirming its validity as the foundation for state-feedback control synthesis.

D. Discretisation and System Bandwidth Analysis

To bridge the gap between continuous mathematical theory and digital microcontroller execution, the linear system was discretised using a zero-order hold (ZOH) transformation. Selecting the appropriate sampling time T_s requires balancing computational load against control degradation caused by digital phase lag.

The selection was driven by evaluating the open-loop system's bandwidth. The highest natural frequency f_{max} of the linearised plant was analytically found via the eigenvalues of the continuous $\mathbf{A}_{lin}$ matrix, yielding $f_{max} \approx 5.04$ Hz. According to the Nyquist-Shannon sampling theorem, the absolute maximum theoretical sampling period to prevent aliasing is $T_{nyquist} \approx 0.1$ s.

While aggressive control heuristics often recommend sampling at $20\times$ the system bandwidth (which would demand a rapid $T_s \approx 0.01$ s), this can be computationally excessive for embedded systems. Therefore, a practical 20 Hz sampling rate ($T_s = 0.05$ s) was selected. This period sits comfortably between the Nyquist limit and the aggressive $20\times$ heuristic, providing a stable control margin while accurately replicating the standard polling capabilities of current, commercially available subsea hardware.

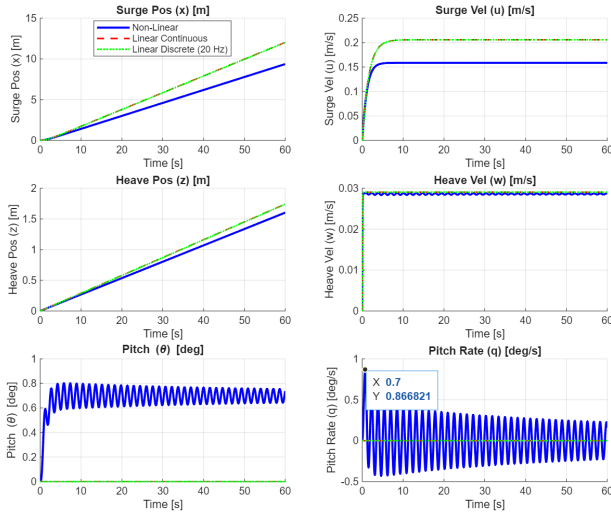


Fig. 3. 3-Way Open Loop Comparison: Non-Linear vs Linear Continuous vs Linear Discrete (20 Hz).

As verified in Fig. 3, the 20 Hz discrete model perfectly overlays the continuous linear baseline without inducing severe phase lag or aliasing. This confirms that 20 Hz provides sufficient fidelity for controller synthesis while remaining computationally lightweight for standard marine electronics.

III. CONTROL SYSTEM DESIGN

The objective of the control system is to command the ROV to track dynamic sequential waypoints in x and z , while maintaining a level pitch ($\theta = 0$).

A. Linear Quadratic Integral (LQI) Formulation

A standard state-feedback regulator (LQR) drives states to zero, but cannot guarantee zero steady-state error in the presence of un-modelled dynamics (such as the quadratic drag intentionally linearised away) or constant environmental disturbances (e.g., ocean currents).

To overcome this fundamental performance limitation, a discrete-time Linear Quadratic Integral (LQI) controller was designed. The standard state vector was augmented with 3 virtual integral states representing the accumulated tracking error for x , z , and θ . The augmented discrete system takes the form:

$$\begin{bmatrix} \mathbf{x}_{k+1} \\ \mathbf{v}_{k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_d & \mathbf{0} \\ -\mathbf{C}_{track}T_s & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}_k \\ \mathbf{v}_k \end{bmatrix} + \begin{bmatrix} \mathbf{B}_d \\ \mathbf{0} \end{bmatrix} \mathbf{u}_k \quad (10)$$

where $\mathbf{v}_k$ is the integral state vector and $\mathbf{C}_{track}$ extracts the positional states.

The optimal feedback gain matrix $\mathbf{K} = [\mathbf{K}_x \quad \mathbf{K}_i]$ was computed by solving the discrete algebraic Riccati equation to minimize the cost function J :

$$J = \sum_{k=0}^{\infty} (\mathbf{x}_{aug,k}^T \mathbf{Q} \mathbf{x}_{aug,k} + \mathbf{u}_k^T \mathbf{R} \mathbf{u}_k) \quad (11)$$

B. Tuning and Design Trade-offs

Engineering judgement was heavily exercised during the tuning of the $\mathbf{Q}$ and $\mathbf{R}$ weighting matrices. The $\mathbf{Q}$ matrix was tuned to heavily penalize positional tracking errors, particularly depth and pitch ($Q_z = 100, Q_\theta = 500$), ensuring tight regulation and rapid disturbance rejection. The integral states were assigned moderate weights to provide sufficient corrective action without inducing low-frequency oscillations.

The control effort penalty $\mathbf{R} = 10\mathbf{I}_{4 \times 4}$ was chosen to deliberately allow aggressive transient responses. Lowering $\mathbf{R}$ further resulted in unacceptable high-frequency thruster chatter, representing a critical trade-off between tracking speed and actuator longevity.

C. Actuator Saturation and Anti-Windup

Physical ROV thrusters are strictly limited; in the simulated environment, they saturate at ± 20 N. When aggressive LQI gains demand thrust beyond these limits, the physical system cannot comply. If left unaddressed, the virtual integral states will continue to accumulate error, leading to massive overshoot and potential instability once the error finally changes sign.

To adapt the standard LQI method beyond taught linear constraints, a back-calculation anti-windup scheme was integrated. If the commanded thrust $\mathbf{u}_{unbounded}$ exceeds $\mathbf{u}_{max}$, the saturated difference is fed back to decrement the integral state dynamically using a back-calculation gain $\mathbf{K}_{aw} = (\mathbf{K}_i^\dagger)/T_s$. This ensures the internal controller states remain synchronized with the physical reality of the actuators.

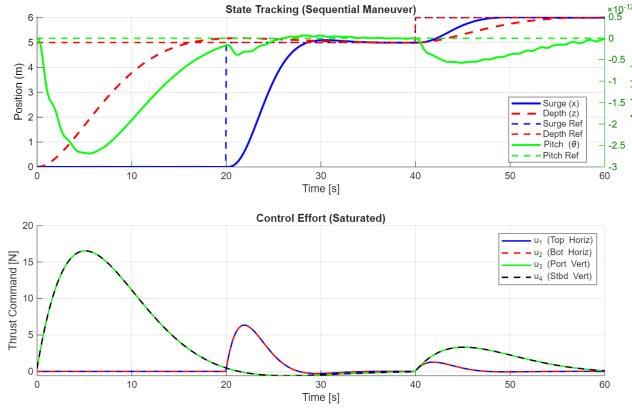


Fig. 4. LQI tracking performance on the idealized Linear Plant.

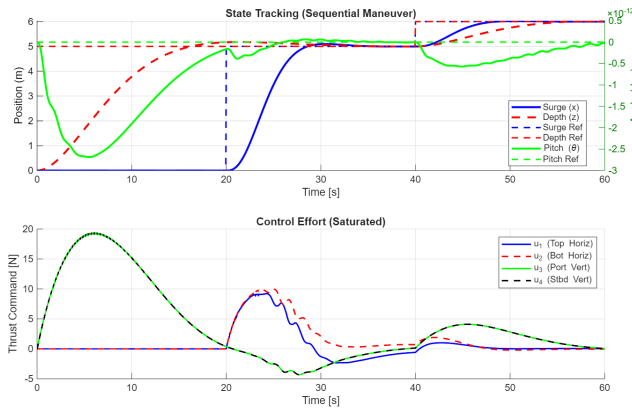


Fig. 5. LQI tracking performance on the true Non-Linear Plant.

D. Performance Analysis: Linear vs. Non-Linear

The LQI controller was first evaluated on the linear plant, and then critically tested against the true, coupled Non-Linear plant.

Comparing Fig. 4 and Fig. 5 provides profound insight into the robustness of the LQI design. On the idealized linear plant, the responses are perfectly crisp. When subjected to the non-linear plant, the introduction of quadratic drag naturally slows the physical response. However, thanks to the integral action, the controller seamlessly compensates for this drag, driving the steady-state error strictly to zero. Notice in the lower pane of Fig. 5 that the thrusters rapidly saturate at 20 N during the initial surge, but the anti-windup scheme prevents any consequent overshoot, validating the advanced control architecture.

E. Robustness and Recovery

To further stress-test the controller, the ROV was initialized with a massive spatial and angular offset ($x = 2\text{m}$, $z = 3\text{m}$, $\theta = 20^\circ$).

As demonstrated in the state responses (Fig. 6), the controller exhibits exceptional resilience. Despite starting completely outside the intended operational envelope, the

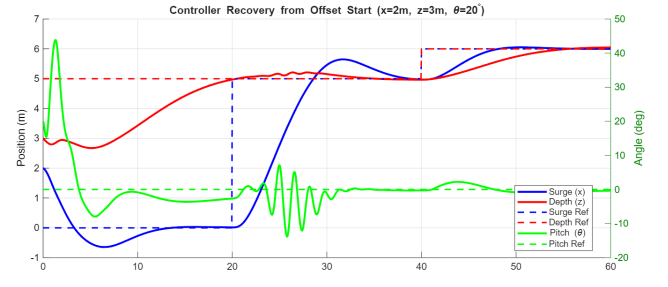


Fig. 6. LQI controller recovery from extreme offset initial conditions.

LQI feedback aggressively saturates the thrusters and safely forces the ROV back to the origin waypoint within 10 seconds, stabilizing the severe 20° pitch offset almost instantaneously.

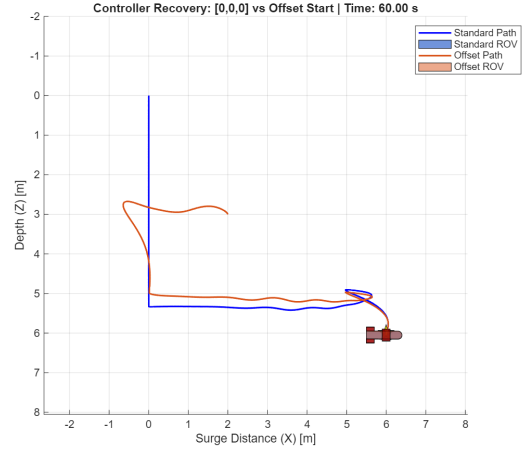


Fig. 7. Spatial trace of the ROV recovering from the $x = 2\text{m}$, $z = 3\text{m}$ offset.

The spatial recovery path is further visualised in Fig. 7. This animated trace confirms that the controller does not merely wildly oscillate back to the origin, but takes a highly coordinated, smooth trajectory through the fluid domain. The vehicle aggressively dives while surging backwards, leveraging its coupled dynamics to arrive precisely at the target coordinate without colliding with the surface or exhibiting unstable looping.

IV. SYSTEM LIMITATIONS AND ENGINEERING TRADE-OFFS

While the formulated state-space model provides an excellent mathematical foundation, the decoupling assumptions and parameters utilized ($m = 16.6\text{kg}$, significant quadratic drag) are inherently characteristic of a heavy, square-profile ROV, primarily drawn from external literature [3]. Conversely, the ideal vehicle for this project is envisioned as a smaller, tube-shaped research ROV. A tubular hull exhibits fundamentally different hydrodynamic properties, specifically reduced form drag and greater natural lateral decoupling. Applying the square-profile parameters to a tubular

design introduces a baseline modelling error, representing a fundamental limitation in the current theoretical approach.

Additionally, the LQI control structure operates under the assumption of full state feedback, requiring real-time measurement of spatial coordinates (x and z). While depth (z) is easily obtained via pressure sensors, surge (x) requires complex external systems like Doppler Velocity Logs (DVLs) or acoustic transponder arrays. Severe environmental limitations, such as confined operational spaces, acoustic multipath interference, or prohibitive hardware costs, often make these measurements unfeasible in practice, breaking the foundational LQI assumption.

A. Blackbox System Mismatch and Robustness Analysis

To evaluate the controller against the aforementioned modelling limitations, it was deployed on a customized “Blackbox” plant. This environment was intentionally designed to represent a third-party attempt at applying the LQI controller to a physically constructed ROV that failed to perfectly replicate the theoretical CAD properties.

Specific un-modelled mismatches were introduced:

- The Center of Gravity (z_g) was shifted from 0.03m to 0.02m, decreasing the hydrostatic restoring moment and rendering the ROV slightly less stable in pitch.
- The vertical thrusters were physically misaligned ($l_{x,v}$ changed from 0 to -0.015 m). This introduced a severe unmodelled pitch moment during any heave commands, violating the core mathematical assumption of the $\mathbf{B}$ matrix.
- The horizontal thrusters were mounted 2cm wider than specified ($l_{z,h}$ increased to ± 0.12 m), artificially altering pitch authority.
- Hull drag coefficients ($X_u, Z_w, X_{u|u}, Z_{w|w}$) were uniformly increased by 15%, simulating a rougher or bulkier hull than the idealized CAD model suggested.

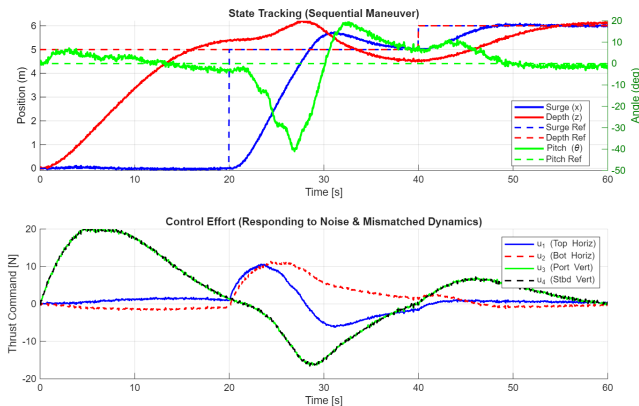


Fig. 8. LQI robustness demonstrated on the noisy Blackbox plant with mismatched dynamics.

The tracking results on the Blackbox plant (Fig. 8) demonstrate that the system remains highly stable, completely overriding the mismatched dynamics. The integral action seamlessly eliminates the unmodelled pitch moments caused

by the misaligned vertical thrusters. However, the control effort exhibits continuous high-frequency chatter as it fights the injected measurement noise. While this chatter significantly increases mechanical wear on the physical thrusters, it represents an acceptable engineering trade-off required to maintain sub-meter tracking accuracy in a harsh, mismatched subsea environment.

V. OBSERVER DESIGN

Because state-feedback control assumes all state variables are perfectly known, unmeasured coordinates like surge (x) must be estimated if acoustic sensors are unavailable. A discrete-time Luenberger Observer was formulated to estimate the full state vector $\hat{\mathbf{x}}$ using only available measurements $\mathbf{y}$:

$$\hat{\mathbf{x}}_{k+1} = \mathbf{A}_d \hat{\mathbf{x}}_k + \mathbf{B}_d \mathbf{u}_k + \mathbf{L}_d (\mathbf{y}_k - \mathbf{C}_d \hat{\mathbf{x}}_k) \quad (12)$$

A. Pole Placement Strategy

The observer gain matrix $\mathbf{L}_d$ determines the convergence dynamics of the estimation error. Rather than manually placing poles, the discrete LQR dual formulation (‘dlqr’) was utilized. This elegant method frames the observer design as an optimal filtering problem, balancing the covariance of process noise (trust in the mathematical model, $\mathbf{Q}_{obs}$) against measurement noise (trust in the sensors, $\mathbf{R}_{obs}$).

The process noise matrix $\mathbf{Q}_{obs}$ was tuned to be exceptionally high, forcing the observer poles deeper into the unit circle than the controller poles. This ensures that the estimation error decays significantly faster than the system dynamics, fulfilling the separation principle and preventing the observer from destabilizing the closed-loop controller.

B. Dead Reckoning & Performance Limitations

A critical limitation was identified during the observability analysis: absolute surge position (x) is mathematically unobservable from internal IMU and pressure data alone. The observer is forced to rely entirely on integrating the estimated velocity u over time, a process known as dead reckoning.

The profound consequences of this limitation are visualized in Fig. 9. When deployed on the Blackbox plant, the observer perfectly tracks pitch and velocities. However, because dead reckoning involves pure integration, any microscopic estimation error, un-modelled disturbance, or sensor noise is continuously accumulated. The estimated x and z states drift linearly away from reality, eventually diverging by several meters.

Dead reckoning is highly not recommended for this type of system and shows why additional measurement is required on these kinds of systems to get x and z (or y). Without this measurement, it is difficult to control these systems. The persistent integration of un-modelled disturbances and noise found in the blackbox environment mathematically guarantees position drift. It is concluded that additional sensory measurement (e.g., Doppler Velocity Logs, Ultra-Short Baseline acoustic transponders, or vision systems) is strictly required to reliably obtain absolute position for control [5]. Without these measurements, it is virtually

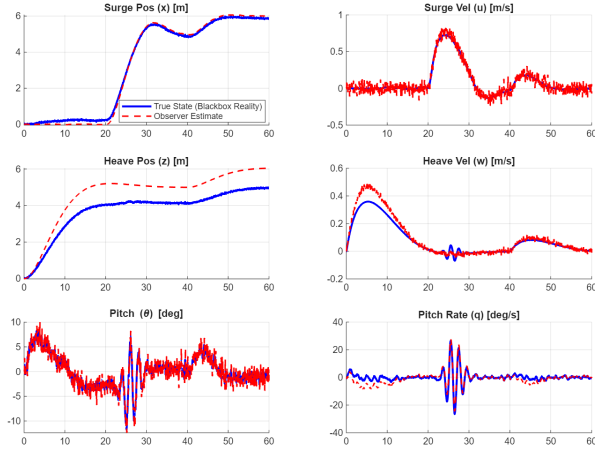


Fig. 9. Luenberger Observer drift due to dead reckoning on the noisy Blackbox plant.

impossible to safely regulate and control the ROV over extended autonomous periods.

VI. CONCLUSION

This report successfully engineered a comprehensive control architecture for an underactuated ROV, explicitly balancing theoretical design against practical subsea limitations. A 3-DOF planar model was derived and validated. A 20 Hz LQI controller augmented with anti-windup provided exceptional, robust waypoint tracking, effortlessly handling actuator saturation and severe mismatched blackbox dynamics. However, implementation of an optimal Luenberger state observer revealed a fatal flaw in the sensor suite: relying on dead reckoning for position coordinates inevitably leads to terminal drift. Future iterations must incorporate physical position sensors to guarantee true autonomous operational safety.

ACKNOWLEDGEMENTS

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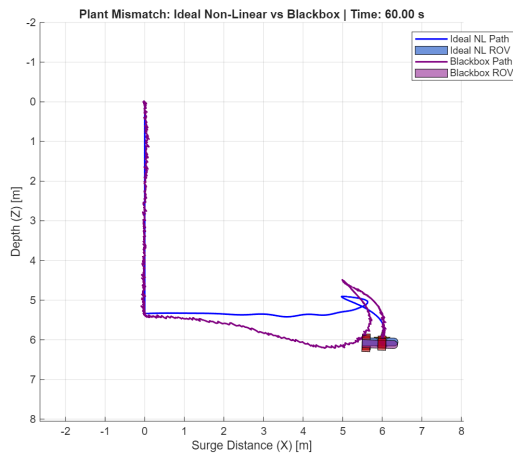


Fig. 10. 2D Animation Snapshot: The ROV tracking the waypoint safely.